MIL-T-48140 (MU)
24 April 1973
SUPERSESSION
(see 6.4)

MILITARY SPECIFICATION

TUNGSTEN POWDER
(FOR PYROTECHNICS)

1. SCOPE

1.1 This specification covers two types of tungsten metal powders for use in pyrotechnic compositions.

1.2 Classification - The tungsten powder shall be of the following types as specified (see 6.2).

Type I - as supplied
Type II - De-agglomerated

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Inspection Terms and Definitions
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

PUBLICATIONS

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Designation E158 - Spectrochemical Computations
ASTM Designation E159 - Hydrogen loss of copper, tungsten and iron powders

FSC: 1376

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ASTM Designation B330 - Method of test for average particle size of refractory metals and compounds by the Fisher subsieve size

ASTM Designation B430 - Particle distribution of tungsten; metal powder by turbidimetry

(Applications for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, Pa., 19103)

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Purity of tungsten powder - The tungsten powder shall have a purity of 99.9 percent minimum, when tested as specified in 4.4.5 (see 6.5).

3.2 Metallic impurities - The maximum percent metallic impurities allowed shall be as specified in Table I when tested as specified in 4.4.1.

TABLE I

Elements	Percent	Elements	Percent
Aluminum	0.002	Manganese	0.001
Cobalt	0.002	Molybdenum	0.040
Chromium	0.004	Nickel	0.001
Copper	0.002	Lead	0.001
Iron	0.025	Silicon	0.003
Magnesium	0.001	Tin	0.001

3.3 Oxygen content - The oxygen content shall be 0.08 percent max, when tested as specified in 4.4.2.

3.4 Average particle size - The average particle size, when tested as specified in 4.4.3, shall be:

a.	Type I	4.2 $\pm$ 0.5 microns
b.	Type II, as supplied	4.2 $\pm$ 0.6 microns
c.	Type II, lab milled (a)	3.9 $\pm$ 0.5 microns

(a) See ASTM Designation B430

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3.5 Particle size distribution - The particle size distribution shall conform to the requirements of Table II when determined as specified in 4.4.4.

TABLE II

Particle Size distribution, microns	Weight Percent					
	Type I Agglomerated As supplied		Type II De-Agglomerated			
			As supplied		Lab milled	
	Min	Max	Min	Max	Min	Max
0 - 5	6	18				
5 - 10	22	38				
10 - 15	20	35				
15 - 20	12	25				
20 - 25	0	25				
0 - 1			0	2	0	5
1 - 2			1	5	2	12
2 - 3			5	14	8	18
3 - 4			10	20	12	24
4 - 5			11	23	12	24
5 - 6			11	21	8	18
6 - 7			8	17	4	12
7 - 25			BALANCE		BALANCE	

3.6 Workmanship - The powder shall be free of all organic impurities, such as: wood chips, dust, floor sweepings and other organic impurities not specified when inspected as specified in 4.4.6.

3.7 First article inspection - This specification makes provisions for first article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform

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any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein.

4.1.1 Submission of product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attest that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Specification number and date, together with identification and date of changes thereto.

c. Certificates of analysis on all materials used directly by the contractor when such materials is controlled by Government specifications shall be made available upon request by the Contracting Officer.

d. Quantity of product in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission.-Prior to the start of regular production the contractor shall submit a first article sample (see 6.2c) to a Government approved facility as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article shall consist of 50 grams of tungsten powder in accordance with instructions issued by the Contracting Officer. All samples submitted shall have been produced by the contractor using the same production processes, procedures, and equipment as will be used in fulfilling the contract. All materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of analysis. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification

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or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspections to be performed.-The sample will be subjected by the Government to any or all of the examinations or tests specified in 4.4. of this specification.

4.2.3 Rejection.-If any sample fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a sample to comply with any of the stated requirements.

4.3 Inspection provisions

4.3.1 Lot formation - A lot shall consist of one or more batches of tungsten powder produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each lot shall consist of that quantity of tungsten powder that has been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous. The product shall be submitted for inspection in accordance with MIL-STD-105 (or MIL-STD-1235 when applicable). The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.2 Testing

4.3.2.1 Sampling - All sampling of individual containers shall be accomplished in such a manner that the sample is representative of the entire contents of the container. Individual or composite samples shall be identified in such a manner as to provide positive identification as to the container or containers represented by each sample. From each lot, a random sample of 10 containers shall be selected. When a lot is comprised of 10 containers or less, each container shall be sampled. Approximately fifty grams of tungsten powder shall be removed from each of the ten containers selected. The samples shall be passed through a sample splitter to composite the samples into two equal size samples. Then the two samples shall be tested in accordance with the test methods of Section 4. If any of the samples fail to meet the requirements specified, the lot shall be rejected.

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Purity of tungsten (see 3.1)	Major Defect	Code No. 05001
Metallic impurities (see 3.2)	Major Defect	Code No. 04001
Oxygen content (see 3.3)	Major Defect	Code No. 02001
Average particle size (see 3.4)	Major Defect	Code No. 03001
Particle size distribution (see 3.5)	Major Defect	Code No. 01001
Workmanship (see 3.6)	Major Defect	Code No. 06001

4.4 Test methods and procedures

4.4.1 Metallic impurities - The metallic impurities shall be determined in accordance with ASTM Designation E158. The arc excitation method shall be used. In an atmosphere of argon and oxygen, the sample shall be dispersed using a Stallwood jet.

4.4.2 Oxygen content - The oxygen content shall be determined in accordance with ASTM Designation E159.

4.4.3 Average particle size - The average particle size shall be determined in accordance with ASTM Designation B330.

4.4.4 Particle size distribution - The particle size distribution shall be determined in accordance with ASTM Designation B430.

4.4.5 Purity of tungsten - The purity of the tungsten shall be determined by difference. The metallic impurities and oxygen content shall be subtracted from 100 percent, the remainder being the percent tungsten. The presence of tungsten, as the only major constituent shall be verified by the use of the spectrograph described in 4.4.1.

4.4.6 Workmanship - A visual examination shall be made on the tungsten powder.

5. PREPARATION FOR DELIVERY

5.1 Level C

5.1.1 Packing - Tungsten powder shall be packed and shipped in such a manner as to insure acceptance by common or other carrier for safe transportation at the lowest rate to the point of delivery.

5.2 Markings - Marking shall be in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use.-This tungsten powder is intended for use in pyrotechnic compositions.

6.2 Ordering date.-Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Type specified in the contract.
c. Provisions for submission of first article samples.
d. Acceptance and description sheets - Acceptance and description sheets shall be prepared for each lot in accordance with MIL-STD-1171.

6.3 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.4 Supersession Data - This specification includes the requirements of Purchase Description PA-PD-3056, dated 26 February 1968.

6.5 The individual metallic impurities and oxygen content requirements shall be as specified. However, the sum of the values of the metallic impurities plus the oxygen content shall not exceed 0.1 percent.

Custodian:
ARMY-MU

Preparing Activity:
ARMY-MU

Project Number: 1376-A-002

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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